

# Welcome

**CS 472/572**  
**Unsupervised Machine Learning**

# Clustering 2

Week-3 -Class -1

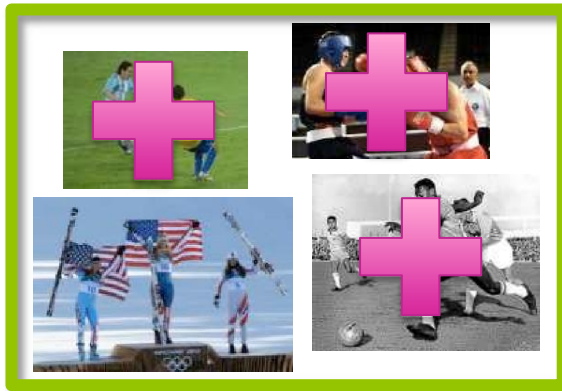
Spring 2026

Instructor: Dr. Khondker Fariha Hossain

# Why a probabilistic approach?

# Learn user preferences

Set of clustered documents read by user



Cluster 1



Cluster 2

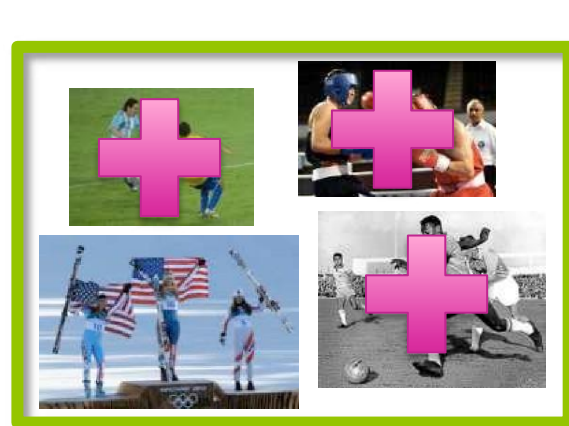


Cluster 3



Use feedback  
to learn user  
preferences  
over topics

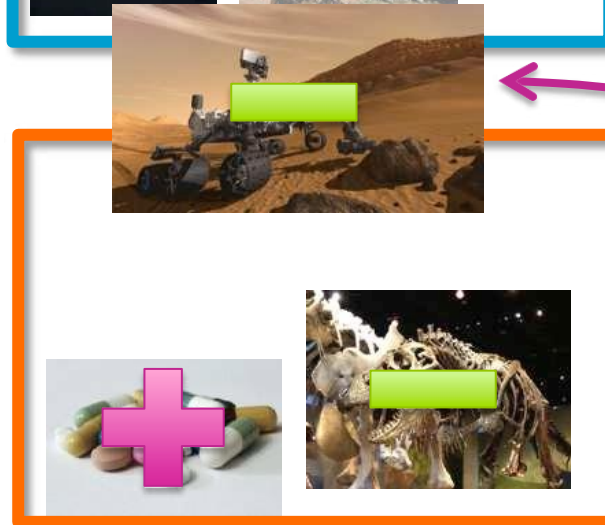
# Uncertainty in cluster assignments



Cluster 1

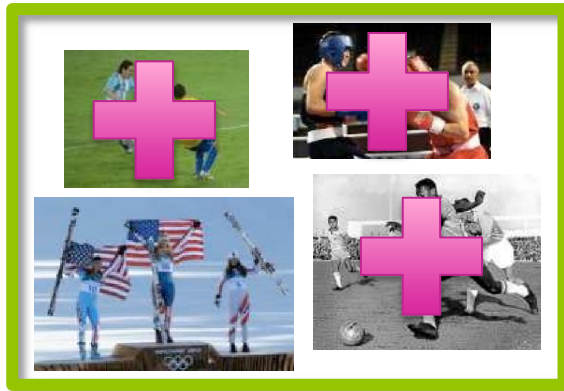


Cluster 3



Slightly closer to  
**Cluster 4** than  
**Cluster 2**, but count  
fully for **Cluster 4**?

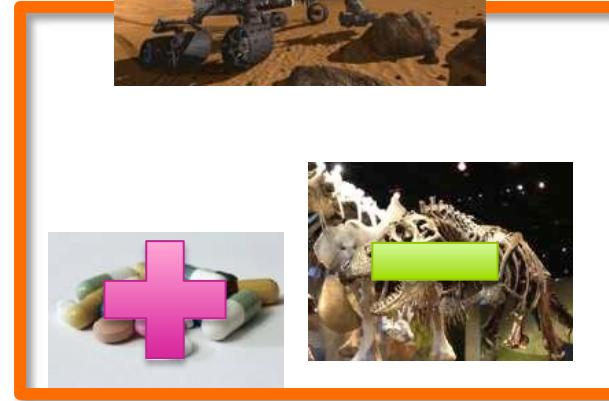
# Uncertainty in cluster assignments



Cluster 1



Cluster 3



Hard assignments  
don't tell full story

# Other limitations of k-means

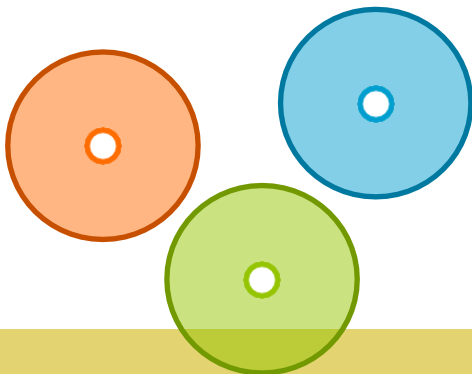
Assign observations to closest cluster center

$$z_i \leftarrow \arg \min_j ||\mu_j - \mathbf{x}_i||_2^2$$

Can use weighted Euclidean,  
but requires *known* weights

Only center matters

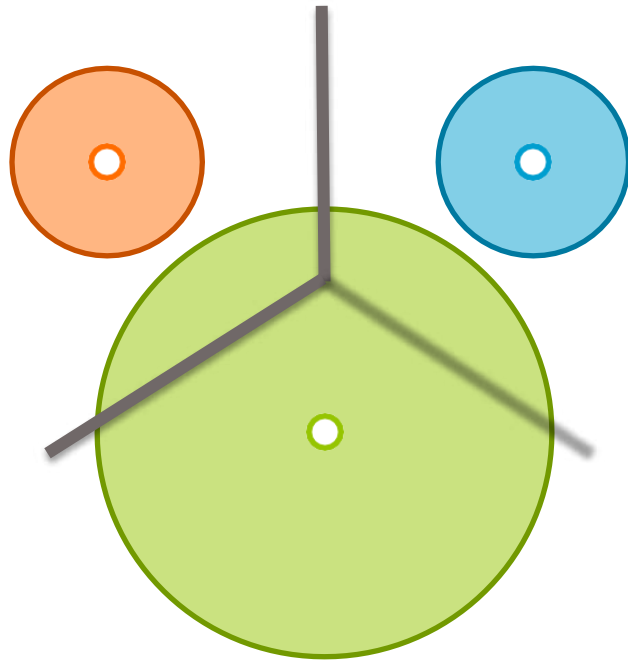
Equivalent to assuming  
*spherically symmetric* clusters



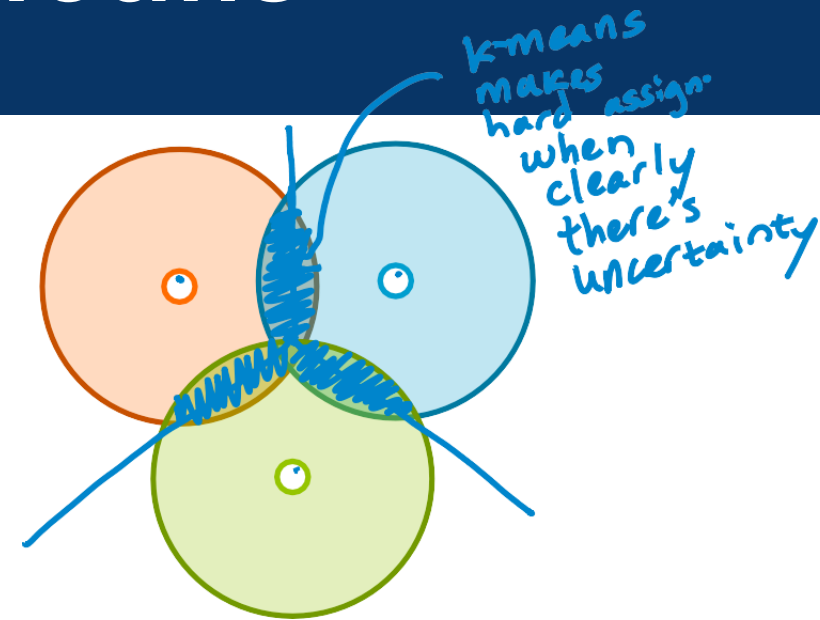
Still assumes all clusters have  
the same axis-aligned ellipses



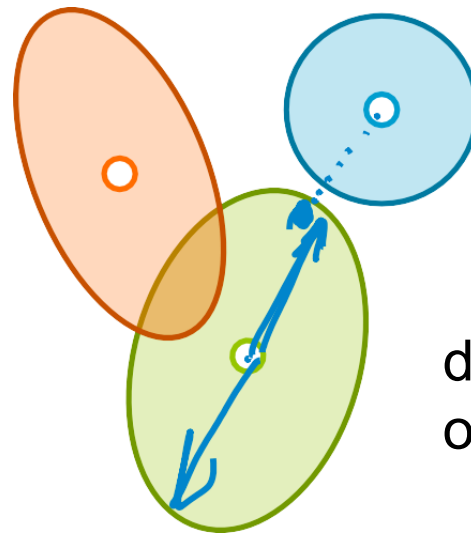
# Failure modes of k-means



disparate cluster sizes



overlapping clusters



different shaped/  
oriented clusters



# Motivates probabilistic model: mixture model

- Provides soft assignments of observations to clusters (uncertainty in assignment)
  - e.g., 54% chance document is world news, 45% science, 1% sports, and 0% entertainment
- Accounts for cluster shapes not just centers
- Enables learning weightings of dimensions
  - e.g., how much to weight each word in the vocabulary when computing cluster assignment

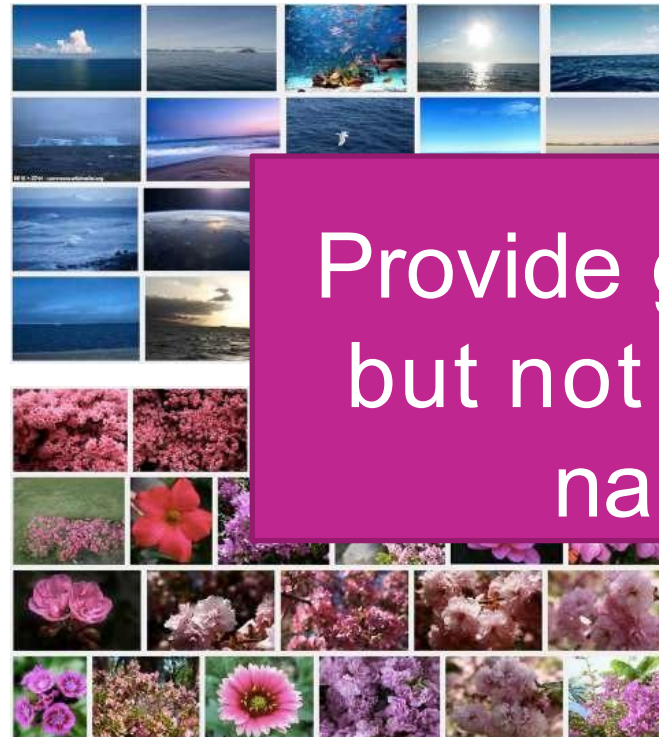


# Mixture models

# Motivating application: Clustering images

Discover groups of similar images

- Ocean
- Pink flower
- Dog
- Sunset
- Clouds
- ...



Provide groupings  
but not category  
names

# Simple image representation

Consider average **red**, **green**, **blue** pixel intensities



[**R** = 0.05, **G** = 0.7, **B** = 0.9]



[**R** = 0.85, **G** = 0.05, **B** = 0.35]



[**R** = 0.02, **G** = 0.95, **B** = 0.4]

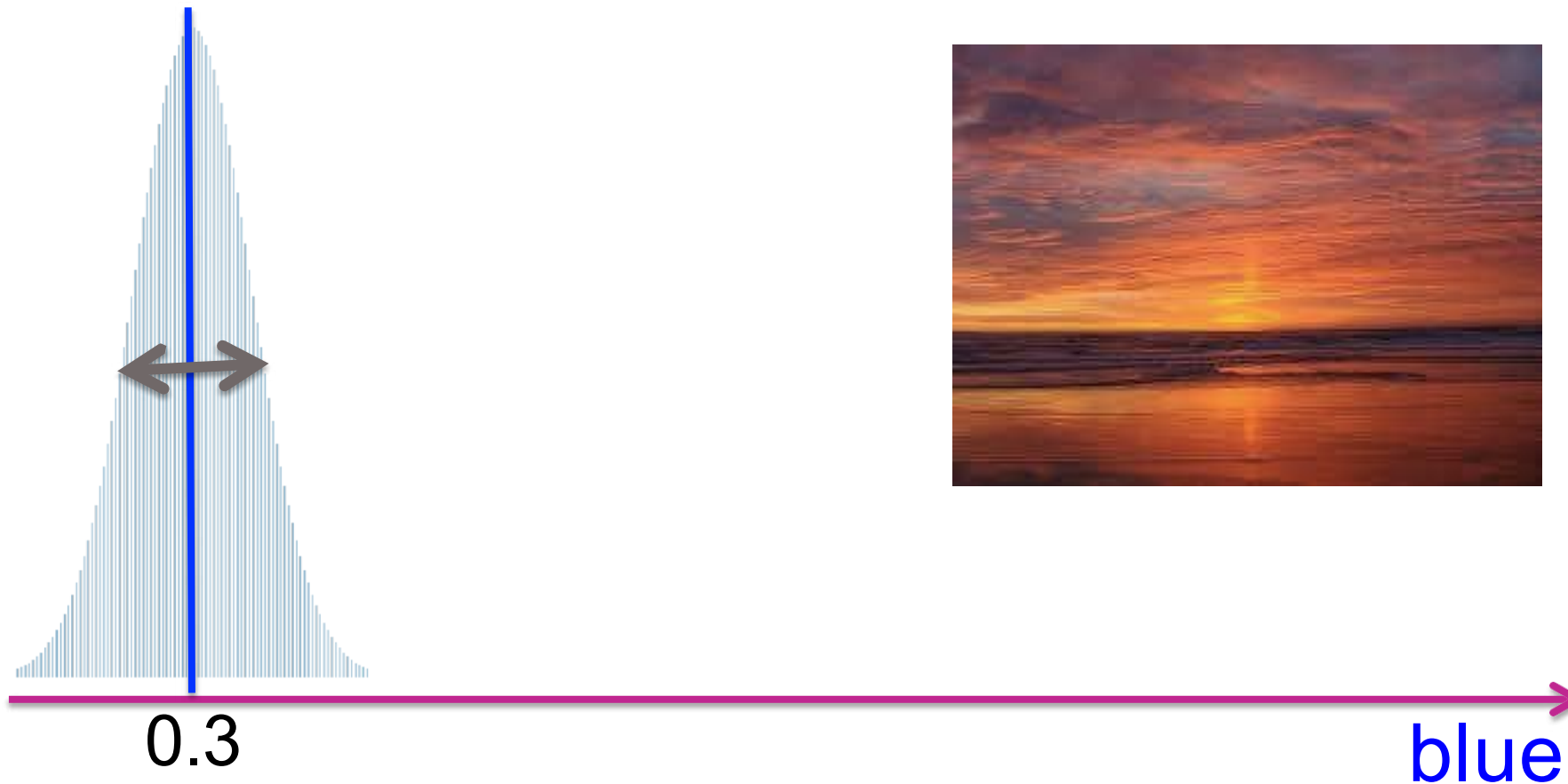
# Distribution over all cloud images

Let's look at just the **blue** dimension



# Distribution over all sunset images

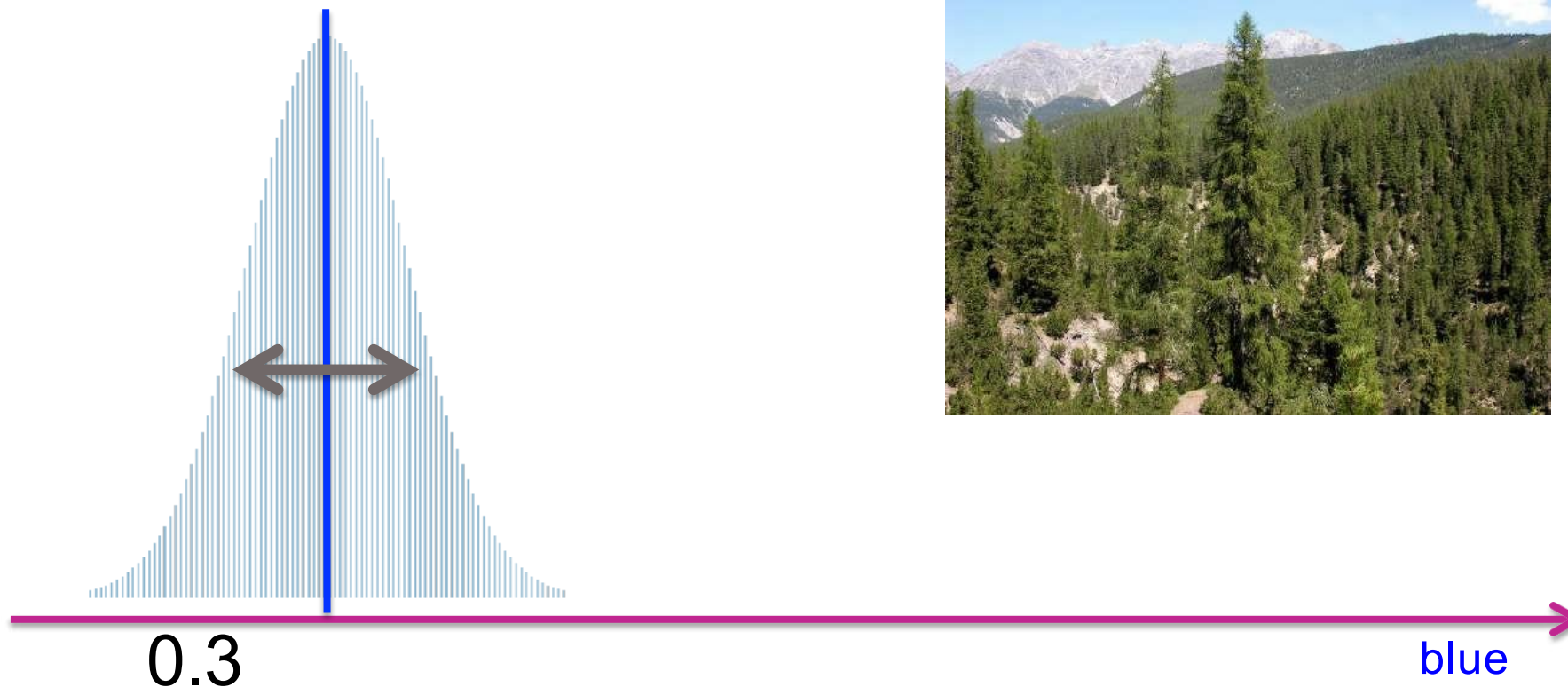
Let's look at just the **blue** dimension



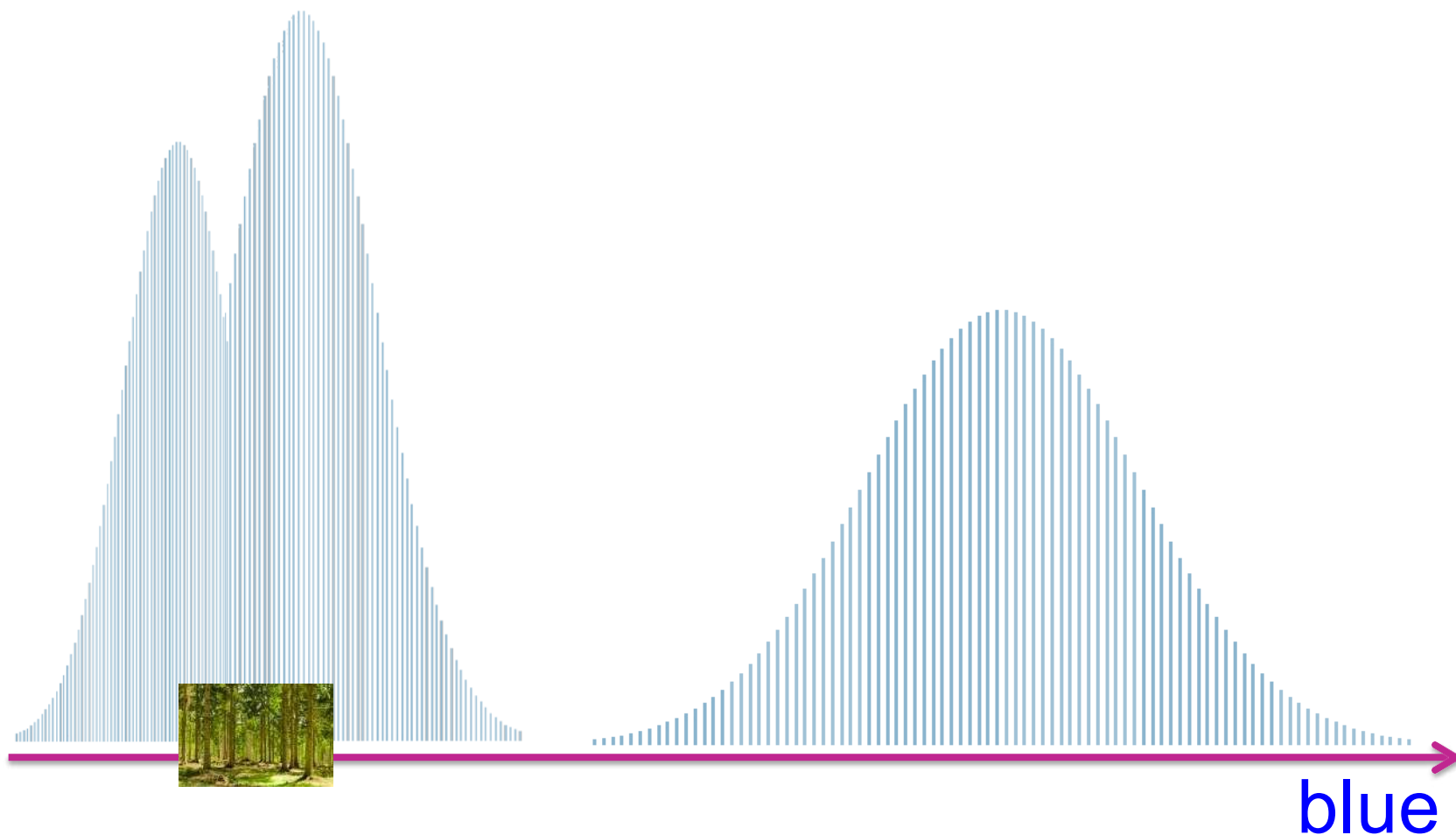


# Distribution over all forest images

Let's look at just the **blue** dimension



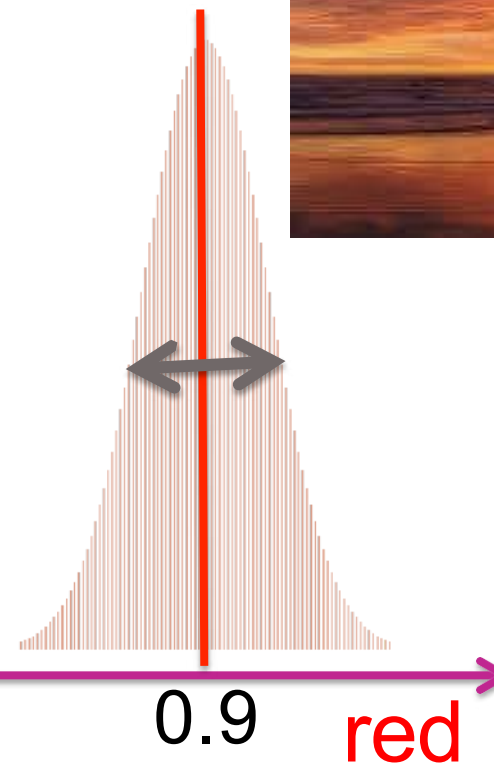
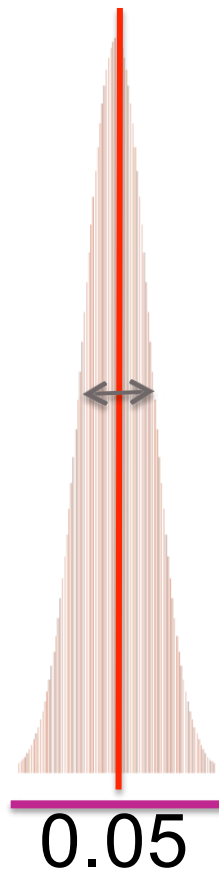
# Distribution over all images





# Can be distinguished along other dim

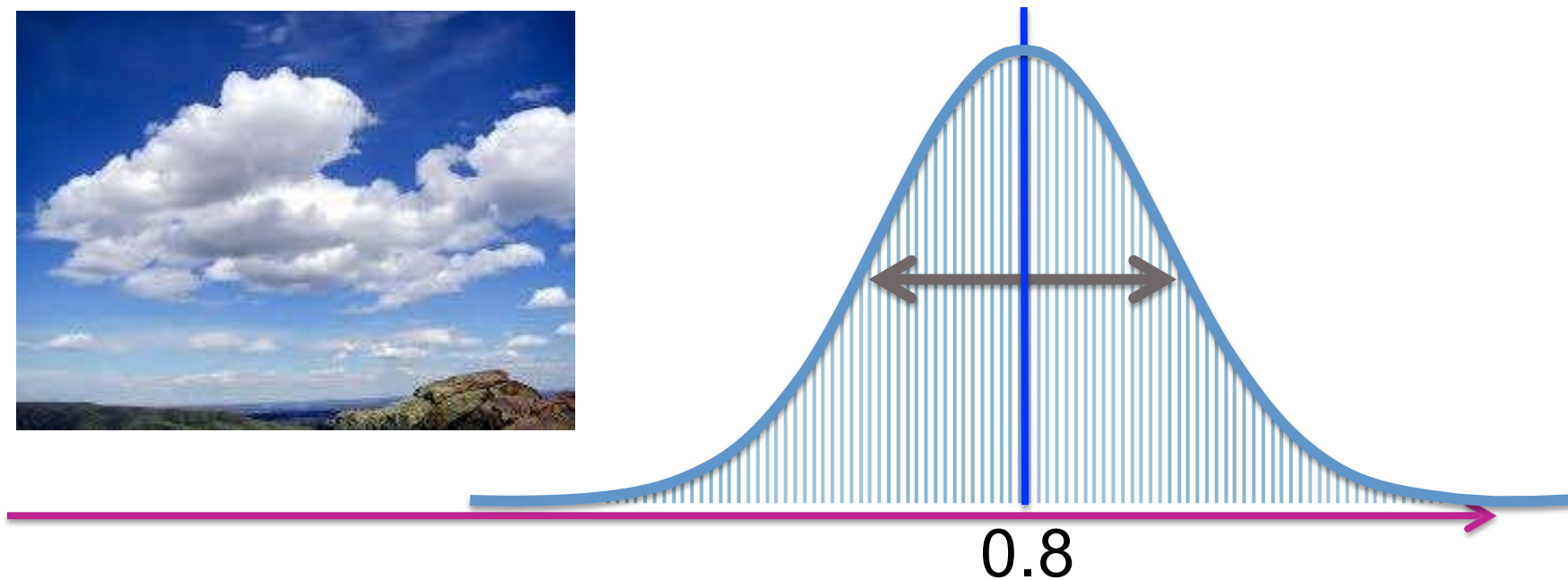
Now look at the **red** dimension



# Background: Gaussian distributions

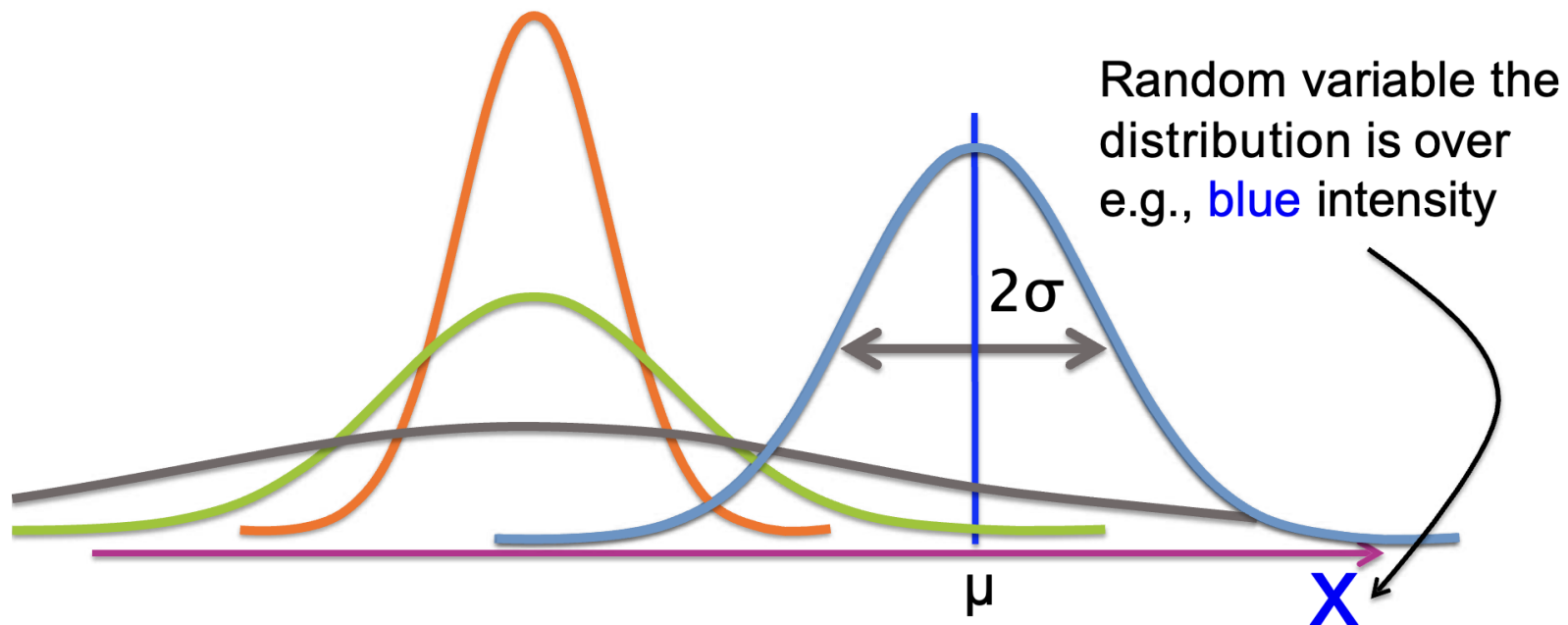
# Model for a given image type

For each dimension of the [R, G, B] vector, and each image type, assume a **Gaussian distribution** over color intensity

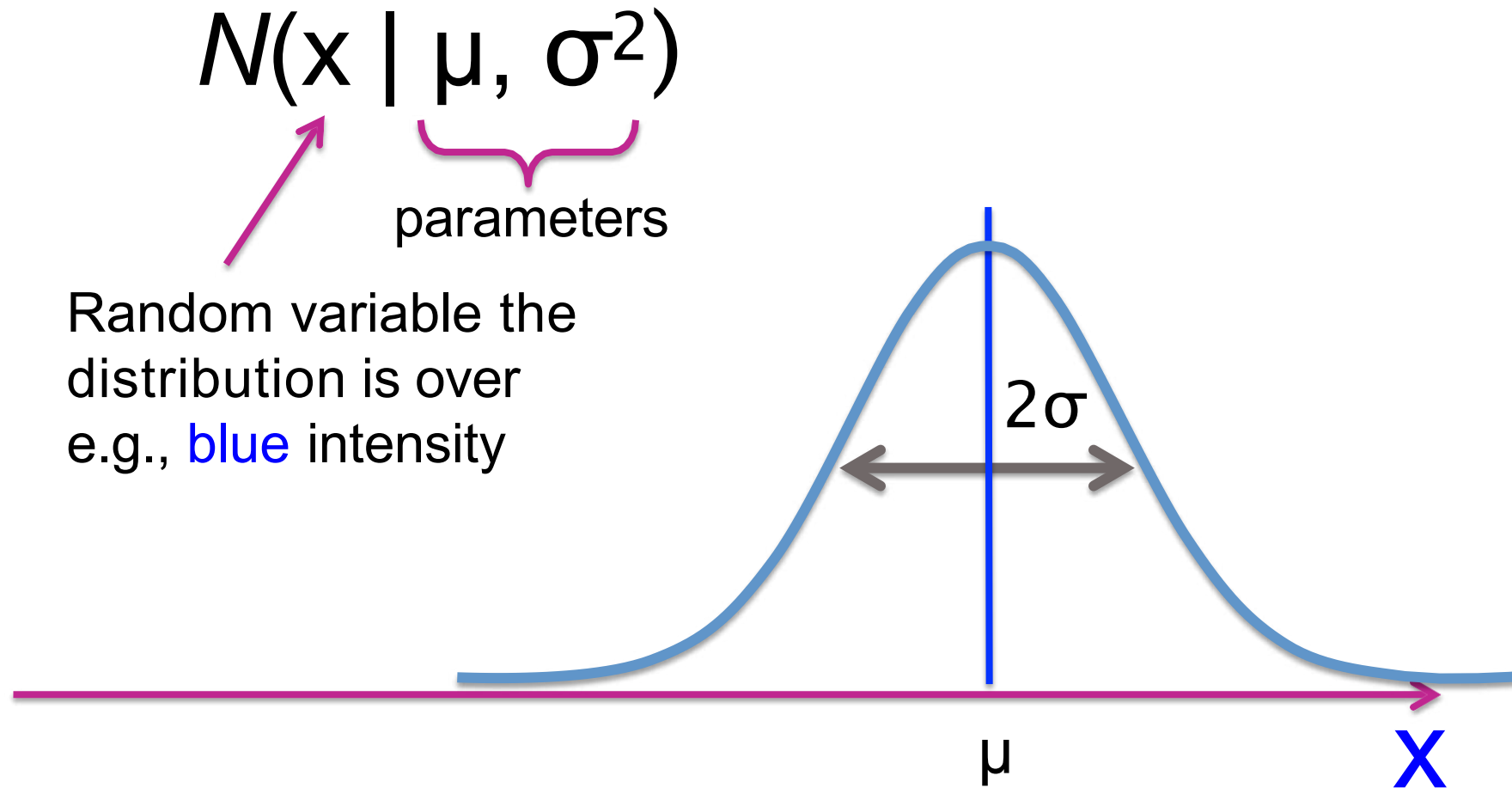


# 1D Gaussians

Fully specified by mean  $\mu$  and variance  $\sigma^2$   
(or standard deviation  $\sigma$ )

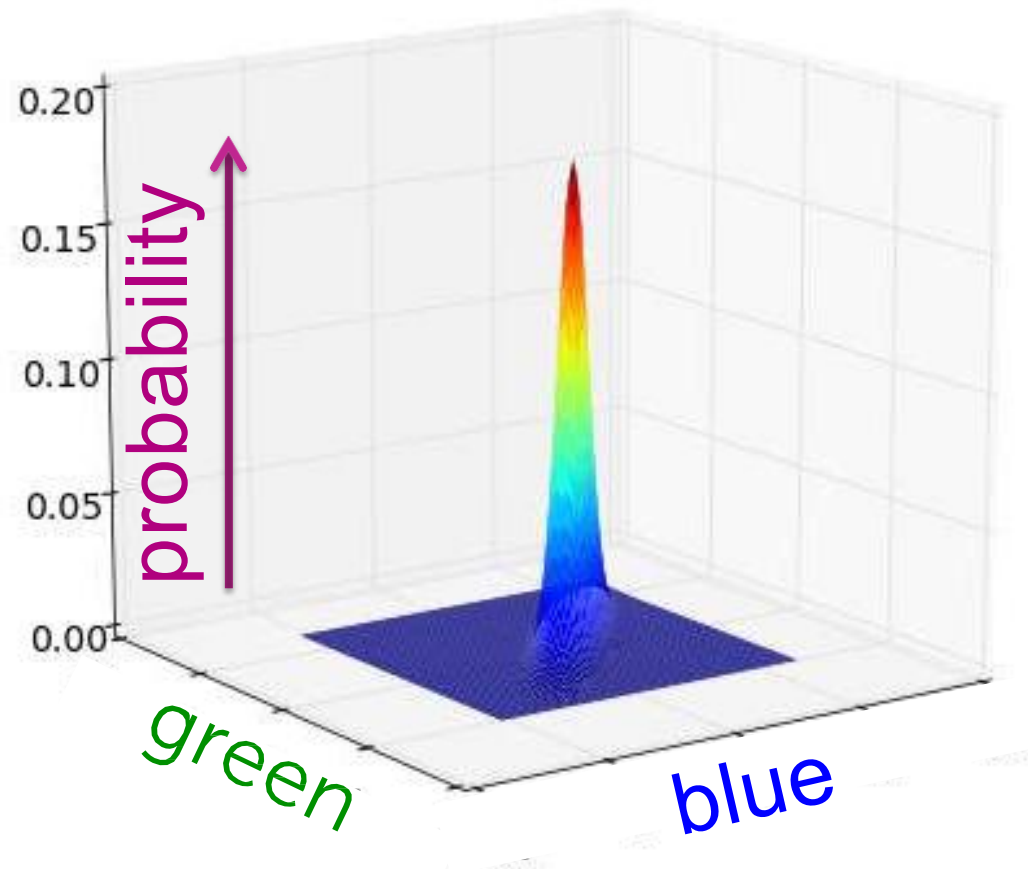


# Notating a 1D Gaussian distribution

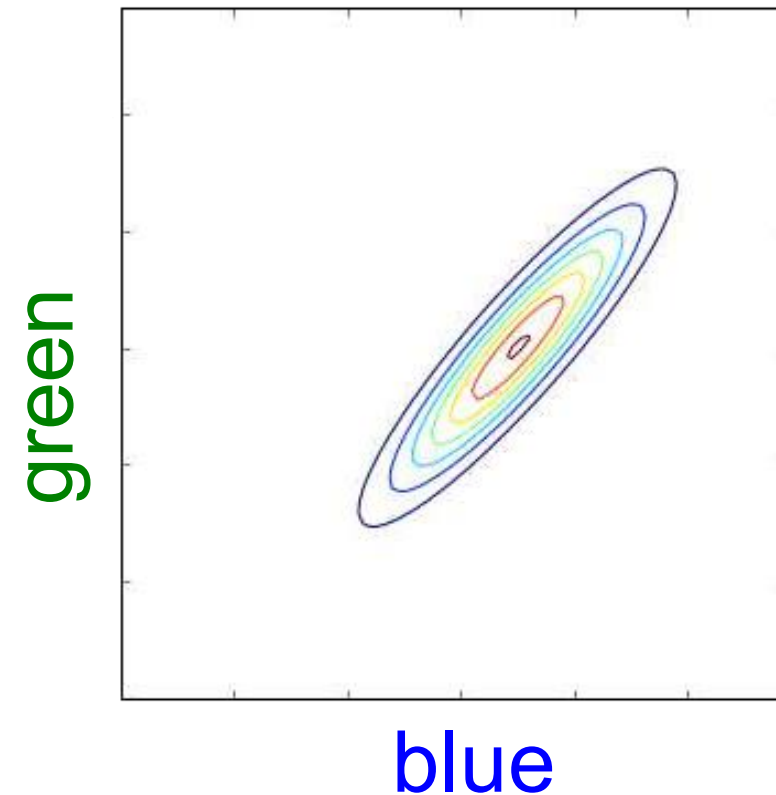


# 2D Gaussians – Bird's eye view

3D mesh plot



Contour plot



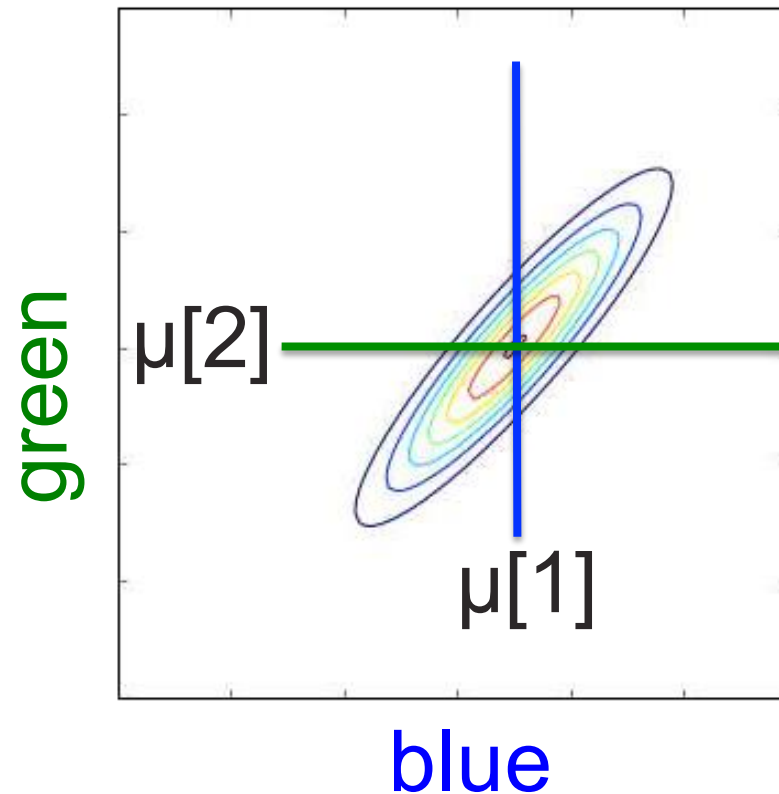


# 2D Gaussians – Parameters

Fully specified by mean  $\mu$  and covariance  $\Sigma$

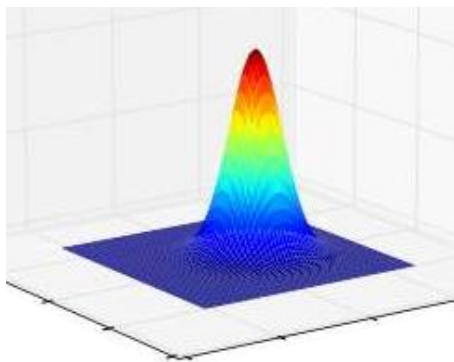
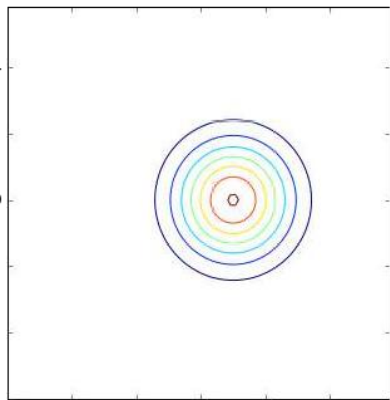
$$\mu = [\mu_{\text{blue}}, \mu_{\text{green}}]$$

mean centers the  
distribution in 2D

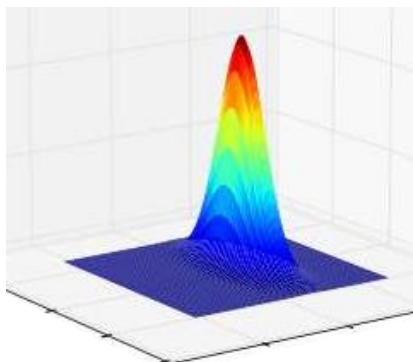
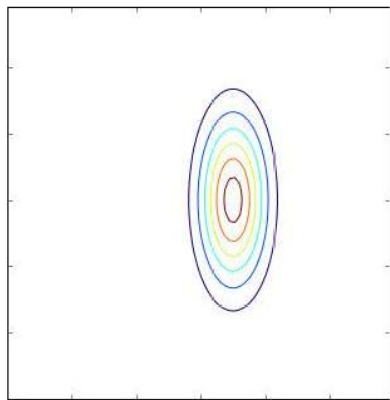


# Covariance structures

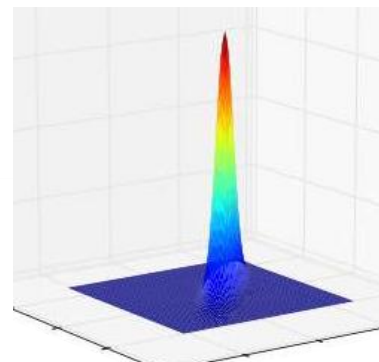
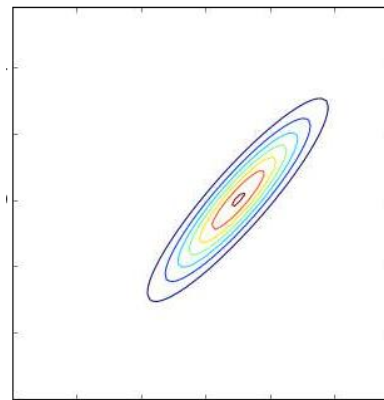
$$\Sigma = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} \sigma_B^2 & 0 \\ 0 & \sigma_G^2 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} \sigma_B^2 & \sigma_{B,G} \\ \sigma_{G,B} & \sigma_G^2 \end{pmatrix}$$

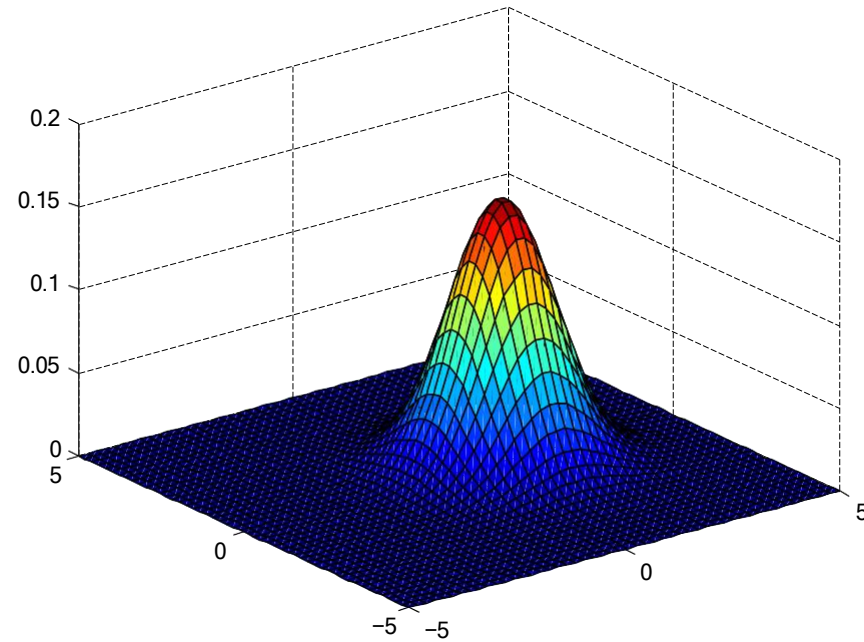


# Notating a multivariate Gaussian

$N(\mathbf{x} \mid \mu, \Sigma)$

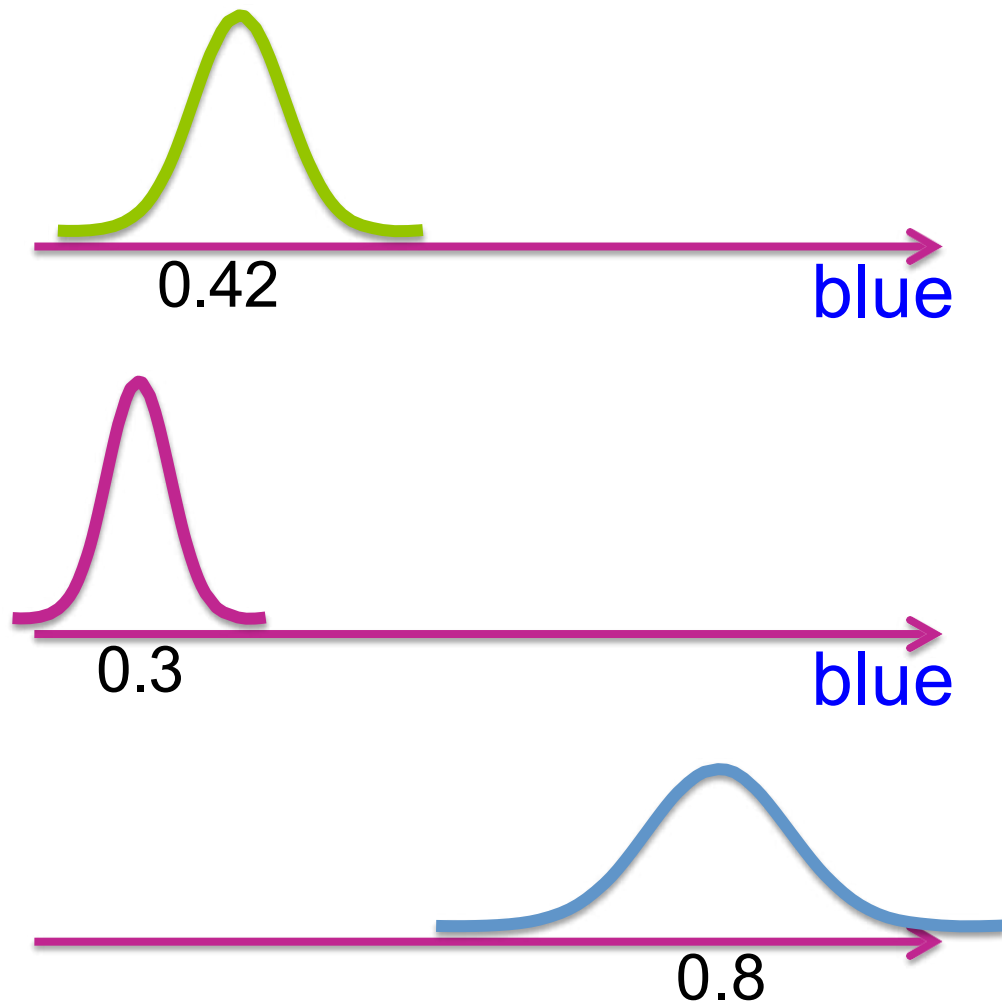
Random **vector**  
e.g., [**R**, **G**, **B**] intensities

**parameters**



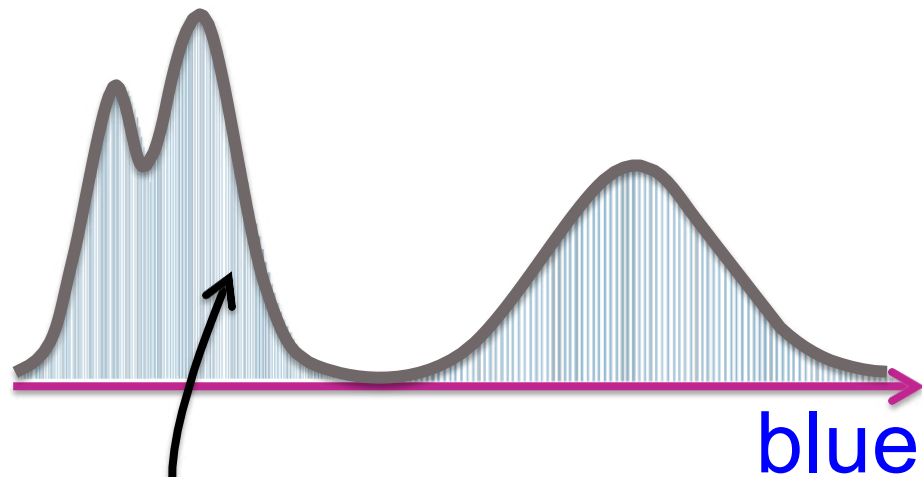
# Mixture of Gaussians

# Model as Gaussian per category/cluster



# Jumble of unlabeled images

HISTOGRAM

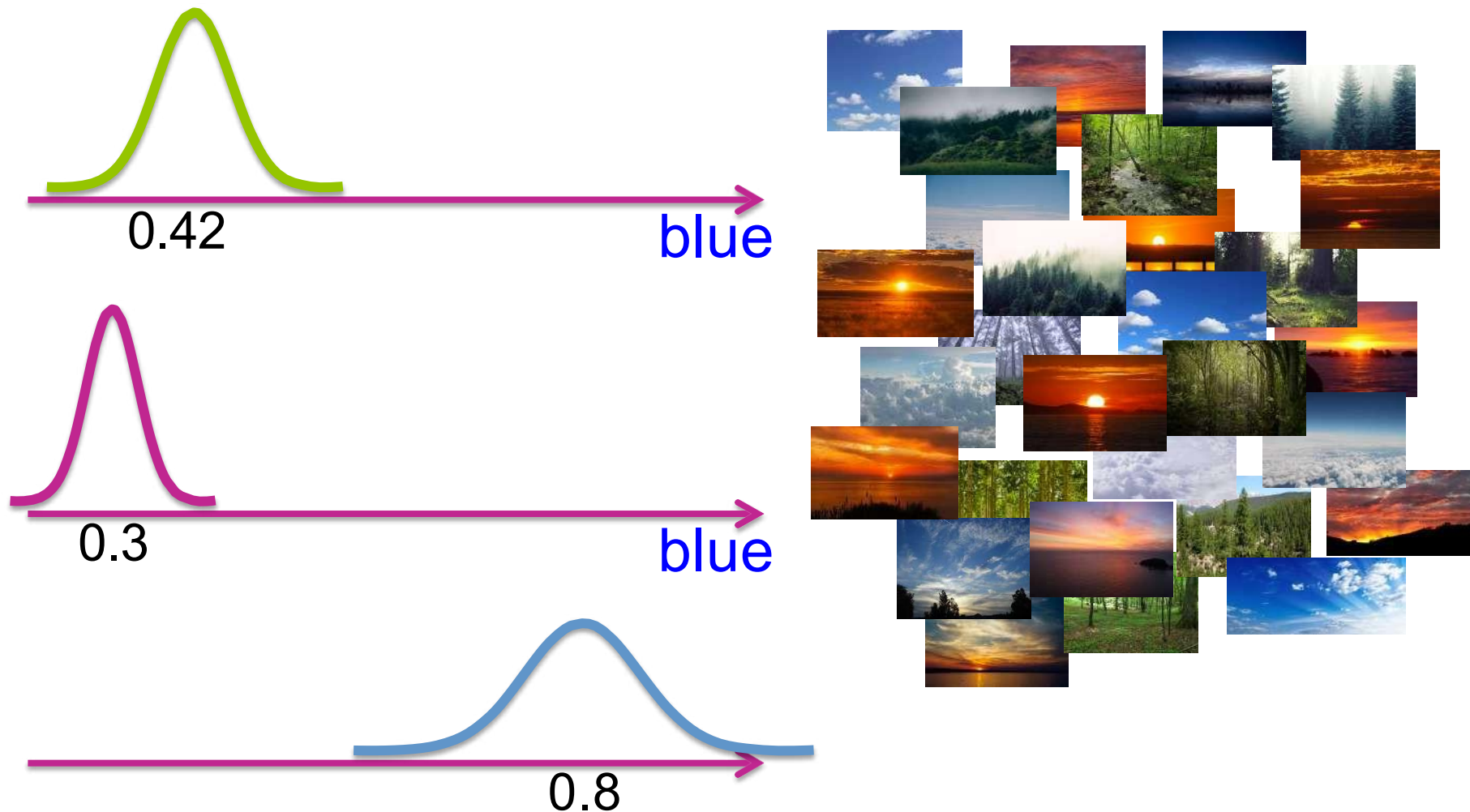


How do we model  
this distribution?





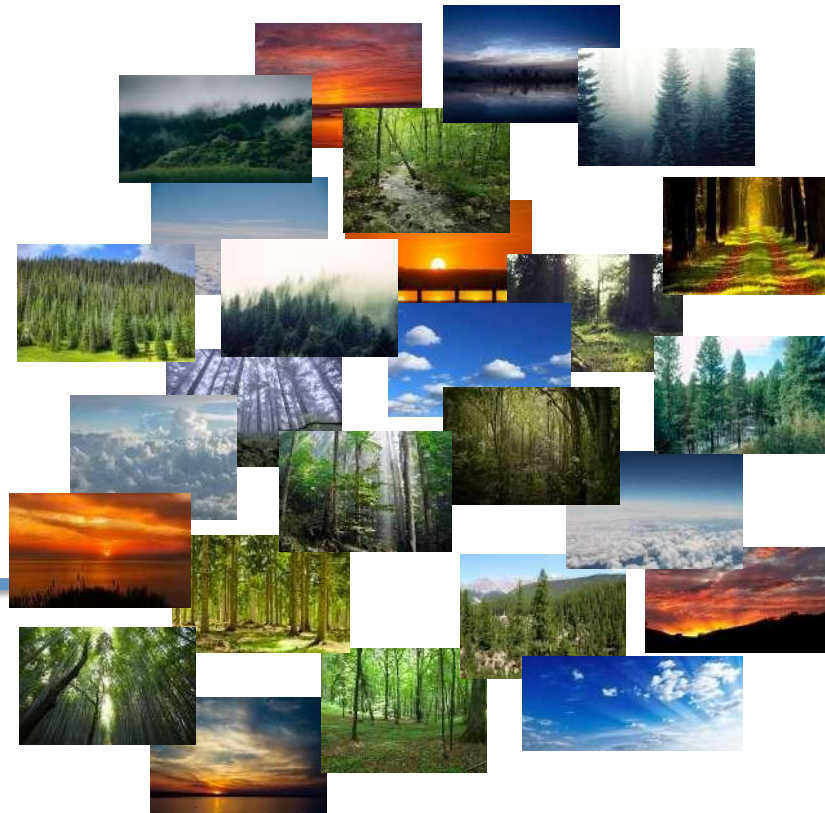
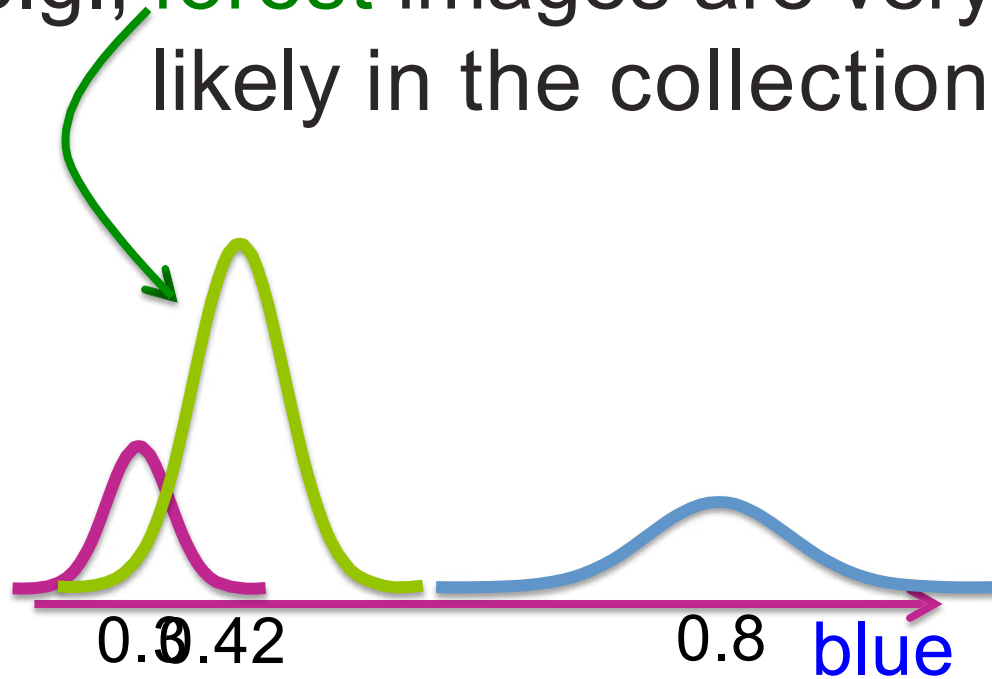
# Model of jumble of unlabeled images





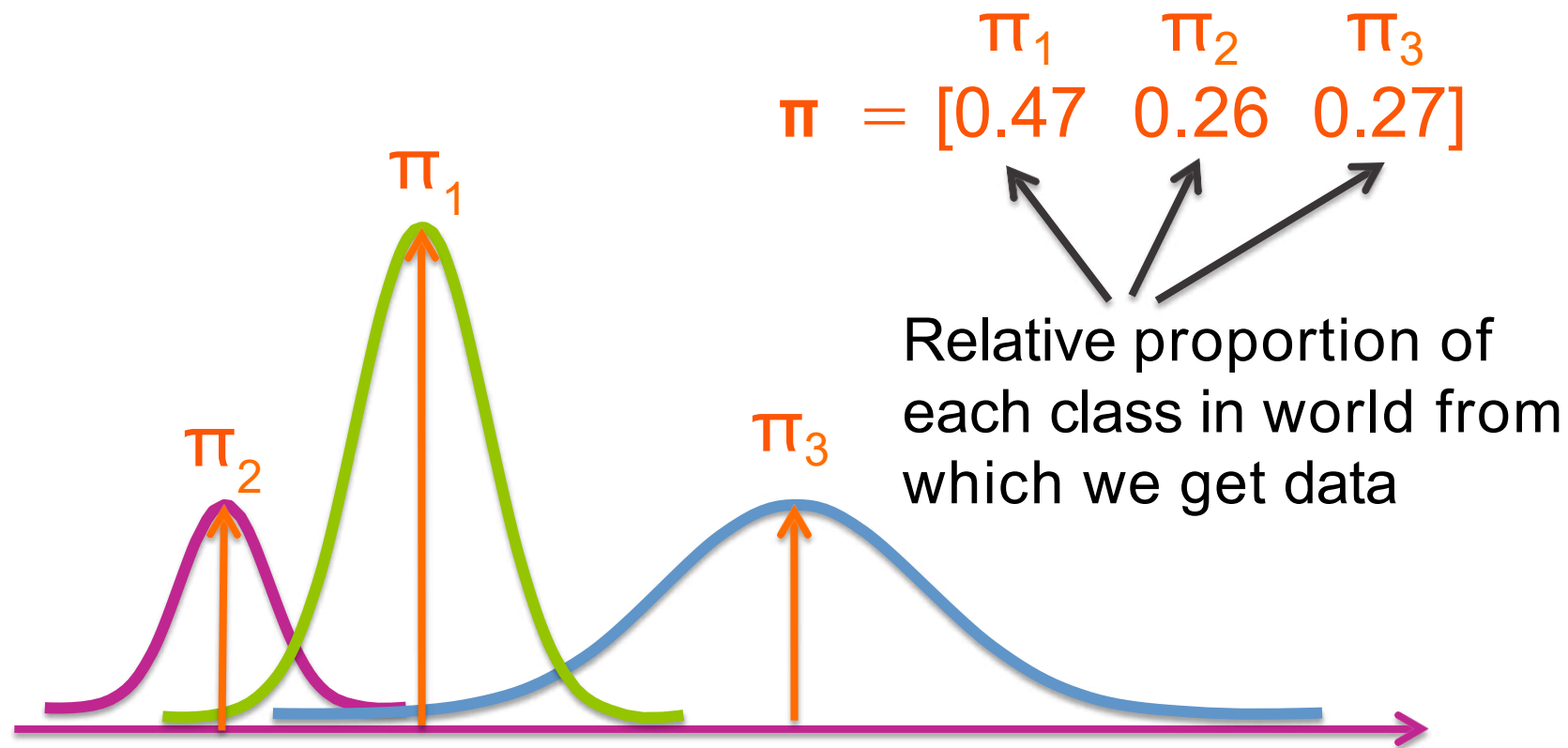
# What if image types not equally represented?

e.g., **forest** images are very likely in the collection



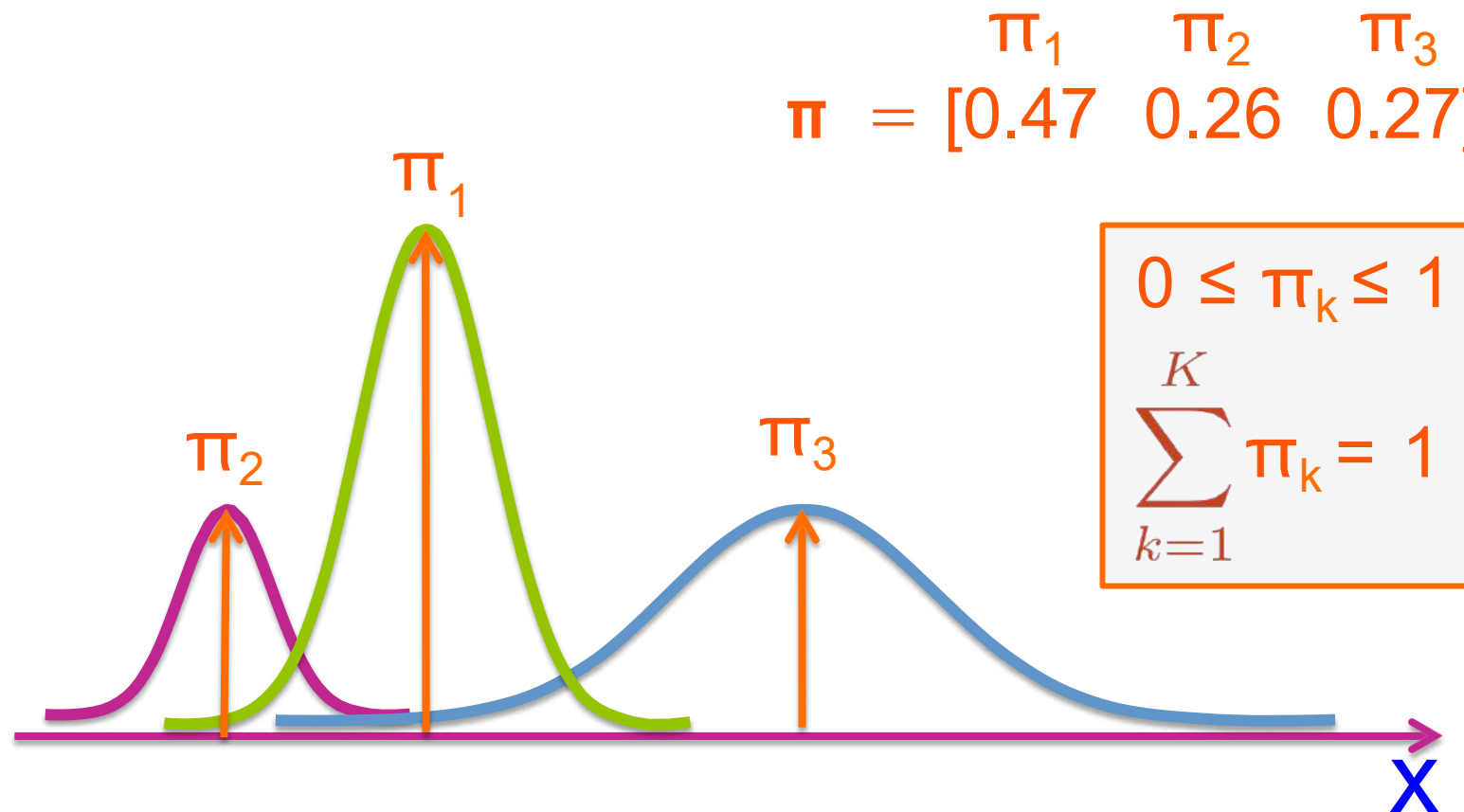
# Combination of weighted Gaussians

Associate a weight  $\pi_k$  with each Gaussian component



# Combination of weighted Gaussians

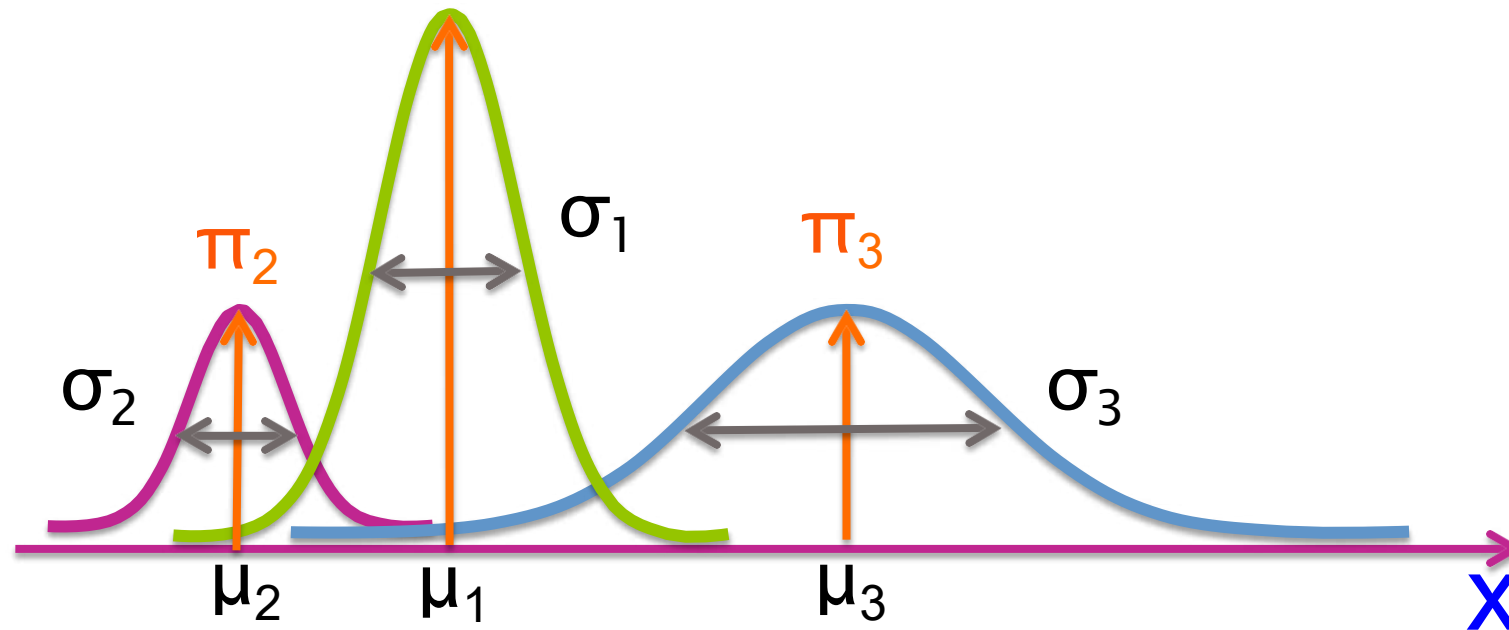
Associate a weight  $\pi_k$  with each Gaussian component



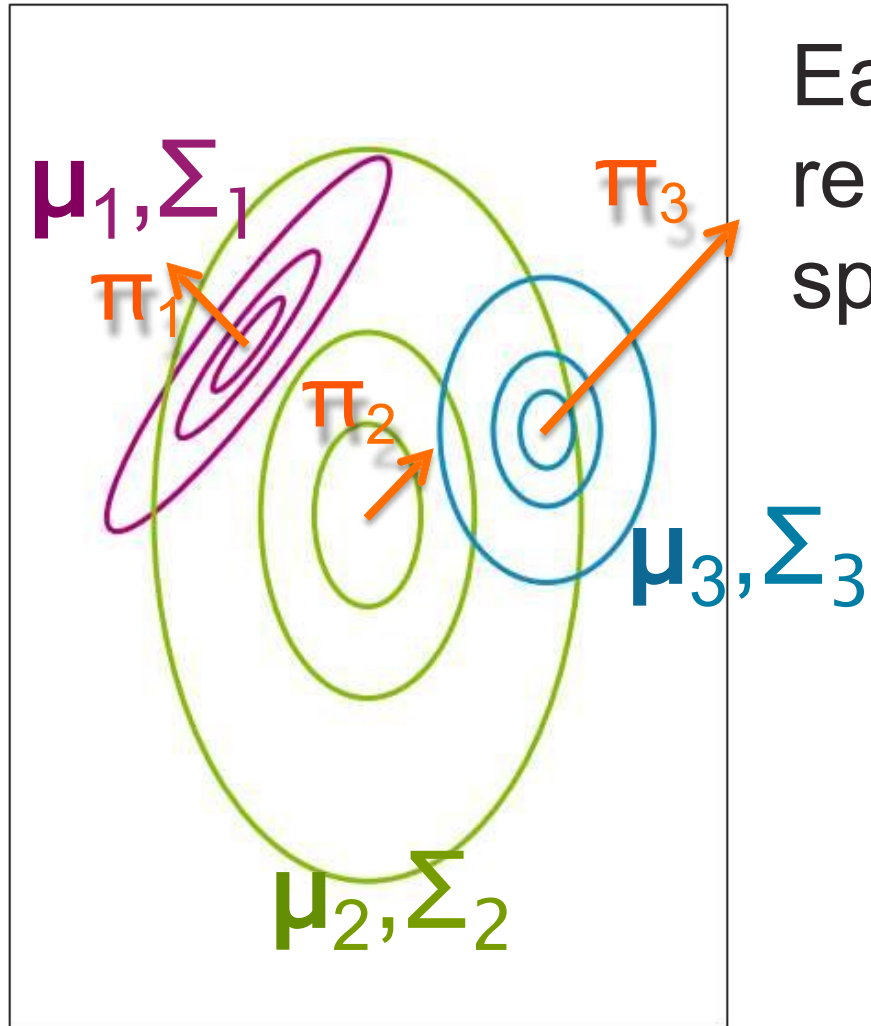
# Mixture of Gaussians (1D)

Each mixture component represents a unique cluster specified by:

$$\{\pi_k, \mu_k, \sigma_k\}^2$$



# Mixture of Gaussians (general)



Each mixture component represents a unique cluster specified by:

$$\{\pi_k, \mu_k, \Sigma_k\}$$

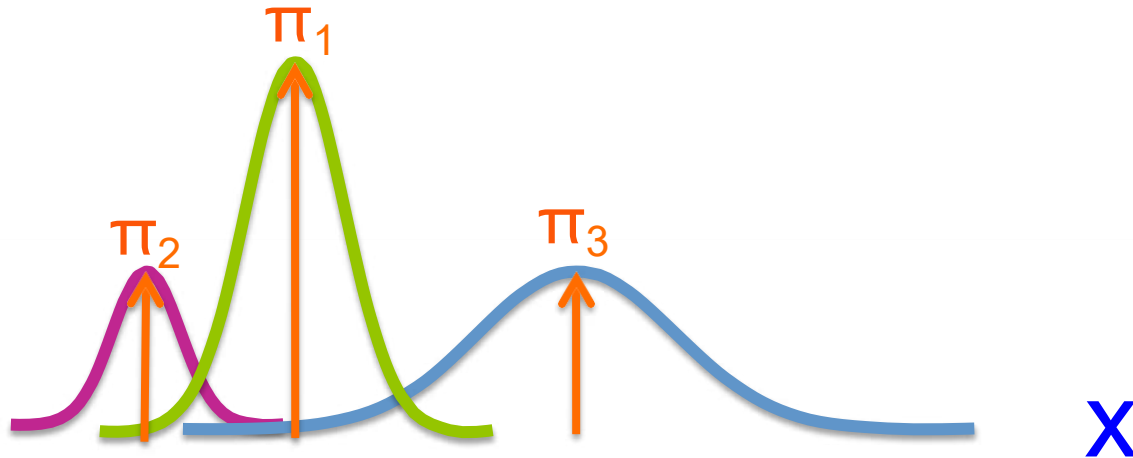
# According to the model...

Without observing the image content, what's the probability it's from cluster  $k$ ? (e.g., prob. of seeing "clouds" image)

$$p(z_i = k) = \hat{\pi}_k$$

Given observation  $x_i$  is from cluster  $k$ , what's the likelihood of seeing  $x_i$ ? (e.g., just look at distribution for "clouds")

$$p(x_i \mid z_i = k, \mu_k, \hat{\sigma}_k) = N(x_i \mid \mu_k, \hat{\sigma}_k)$$



# Thank you

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